

Student Research Topic Guide (UMBL Edition)

This guide is intended for students seeking topics for undergraduate theses, course papers, or independent research projects.

UMBL (Universal Meaning-Based Language) is an early-stage **language-design development project**, exploring how to construct a structured, evolvable pictographic writing language based on modular semantic components.

UMBL is not yet a mature language; it is a **language engineering prototype** suitable for small-scale, controlled, and verifiable academic studies.

This guide proposes possible research directions without claiming any definitive theoretical advantages for UMBL.

1. Introduction: Positioning UMBL as a Research Object

The core goal of UMBL is to explore a **pictographic writing language prototype**, including its character-formation principles, semantic structures, potential grammar, and mechanisms for language evolution.

UMBL is **not** an icon set or UI design project. It is a **language-engineering experiment** that aims, in the long term, to develop into a fully functional constructed language—similar in ambition to Esperanto, but pictographic and evolvable.

The design motivation for UMBL arises from long-term comparative observations of Chinese and English, including:

- Chinese character formation, semantic radicals, and visual transparency
- English reliance on roots and affixes for abstract concept formation
- The strong visual-semantic tendencies in Chinese media culture (subtitles, danmu overlays)
- Limitations of phonetic writing systems in cross-cultural comprehension
- Inconsistencies in natural languages regarding abstraction and logical transparency

These observations motivate the question:

Is it possible to design a structured, compositional, and evolvable pictographic language that offers new possibilities for semantic transparency and cross-cultural understanding?

Because UMBL is open-source, structurally clear, and easy to access, it is highly suitable as a research object for undergraduate students.

2. Possible Research Directions (By Discipline)

2.1 Linguistics (Core Area)

- Analysis of UMBL's character-formation principles
- Structural study of pictographic morphemes
- Combination logic of pictographic words
- Exploration of preliminary grammar prototypes

- Structural comparison between UMBL and Chinese character components
- Comparison with constructed language (conlang) design methodologies
- Feasibility analysis of UMBL's evolution rules

2.2 Cognitive Science

- Learning curves for pictographic languages
- Recognition speed and cognitive load of pictographic symbols
- Misinterpretation patterns in visual semantic structures
- Processing differences between pictographic and linear textual expressions

2.3 Information Design & Standardization

- Analysis of UMBL's modular component design
- Structural comparison with ISO graphical symbols
- Evaluation of visual consistency and readability
- Maintainability and extensibility of character-formation rules

2.4 Computer Science

- Encoding schemes and data structures for UMBL
- Automated composition of pictographic expressions
- Machine-readable pictographic language structures
- Prototype editors or rendering tools for UMBL

2.5 Design Studies

- Experimental design of pictographic words based on UMBL
- Methods for ensuring visual-semantic consistency
- Designer experience studies on constructing pictographic expressions

3. Feasible Research Methods (Undergraduate-Friendly)

3.1 Literature Review

- Constructed language (conlang) research
- Pictographic writing and graphical symbol studies
- Cognitive load and visual processing literature
- Information visualization and standardization research

3.2 Small-Scale User Studies

- Recognition speed
- Comprehension accuracy
- Misinterpretation rates
- Learning curves
- Comparative experiments: pictographic morphemes vs. linear text

3.3 Comparative Analysis

- Structural comparison with Chinese character components
- Semantic transparency comparison with emoji or ISO pictograms
- Structural comparison with other constructed languages (e.g., Esperanto, Lojban)

3.4 Prototype Construction

- Building a small set of pictographic words
- Designing simple pictographic sentence-expression tasks
- Testing semantic transparency and comprehensibility

3.5 Corpus-Style Quantitative Analysis

- Statistical analysis of UMBL component structures
- Frequency and distribution of component combinations
- Semantic domain coverage analysis

4. Citable Materials

- UMBL GitHub repository (component library, whitepaper, design guidelines)
- Literature on constructed languages and language engineering
- Research on pictographic writing and graphical symbols
- Cognitive science and visual language studies
- Public summaries of ISO graphical symbol standards

Students should distinguish between **engineering project materials** and **peer-reviewed academic literature**.

5. Important Notes

- UMBL is not a mature language; it is a language-design prototype.
 - Grammar, phonology, and character-formation rules are still exploratory.
 - Research should focus on **analysis**, **comparison**, and **exploration**, not on “proving UMBL’s superiority.”
 - Research scope should be small and well-defined.
 - Limitations must be clearly acknowledged.
 - User studies must follow basic ethical guidelines.
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This guide aims to provide a practical starting point for student research.

The value of UMBL lies not in offering definitive conclusions, but in serving as a **research-ready, analyzable, and discussable language-design prototype** that opens rich possibilities across linguistics, cognitive science, information design, and computer science.